

Nassau Candy Distributor — Profitability Analysis

End-to-end analytics case study: Excel → SQL → Power BI producing a one-page executive dashboard. Prepared by: Sonu Kumar.



Executive Summary

Total Sales

\$141,783.63

Total Cost

\$48,340.83

Gross Profit

\$93,442.80

Units Sold

38,654

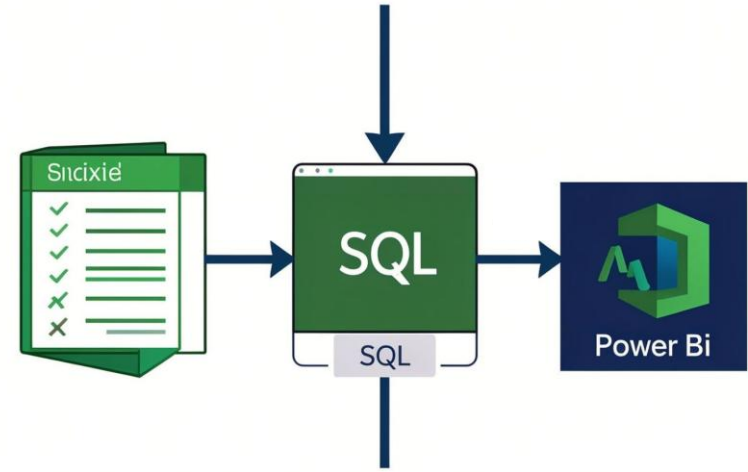
Profit Margin

65.91%

Project Objective & Workflow

Objective: Convert transactional records into a management-focused view of profitability to identify divisions, regions, and products driving gross profit.

Workflow: Excel → SQL → Power BI → Interactive one-page executive dashboard.





Key Questions Addressed

01

Overall KPIs

Sales, Cost, Gross Profit, Units, Margin

02

Division & Region

Which generate the most gross profit?

03

Top Products

Top 5 contributors by gross profit and their margins

04

Interactivity

Filters: Order Date, Region, Division

Dataset Overview & Measures

10,194 transaction records (02 Jan 2024 – 31 Dec 2025), 18 columns. Key measures: Total Orders 8,549; Sales \$141,783.63; Cost \$48,340.83; Gross Profit \$93,442.80; Units 38,654; Margin 65.91%.



Dimensions

Product, Division, Region, Customer, Order Date



Financial Fields

Sales, Cost, Gross Profit

Excel & SQL: Preparation and Validation

大カ教席：ブガウテン回が純：

ブガウテン回が純	Pivot		業務		業務	
	大カ	本カの純	業務	放ベ席	業務	業務
1 スカカダ	✓	✓ワリフック		\$%	※ 業務の角助	
2 業務経理		✓ドゥ日		1%		
3 水カお局	✓		✓✓	✓	0	✓
4 ククト	✓		✓✓	✓	-0	✓
5 パール	✓		✓✓	✓	19	✓
6 ルガラテ			✓✓	✓	40	19
7 プル西書カ	✓	\$ %				
4 業務経	✓	パハ	×Hレゾ	✓ド	×ラ	
5 業務経		ロリカ		✓✓		✓
8 フバワヤッス	✓	✓✓		✓✓		✓
10 パリス	✓	✓✓				
10 パルン	✓	✓✓				
14 大カロインシ			✓	✓		
18			✓	✓		
12 同前ズ	✓					
17 業務経	✓	\$90	✓	✓		\$3
18	✓		レタフ	✓	フラクケンバ	\$
18 (バクツク)	✓		✓	✓	フサダ	10
10 スラン		✓0	✓	✓	ヨバレケービヤン	
18 オカド		✓				
10		✓\$18				
22 ボールジシ						
10						

Excel

Initial inspection, type checks, baseline totals, date parsing (day-first), and quick pivots.

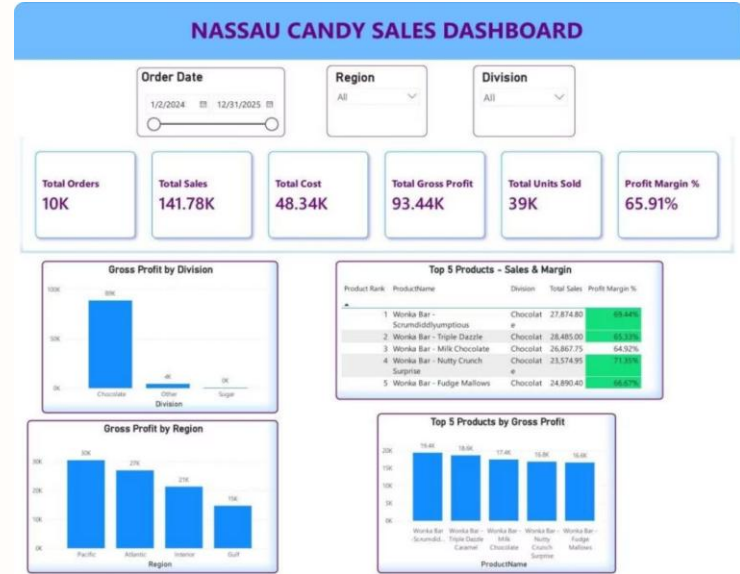
SQL

Load, aggregate, rank and validate results with repeatable queries; reconciled to Excel totals.

Power BI — Dashboard Design

One-page executive layout: header, slicers (Order Date, Region, Division), KPI strip (6 cards), analysis area with gross-profit by division, region, and Top 5 products (sales & margin).

Design principle: each visual must support a business decision; preserve white space for readability.



Top Findings — Profit Concentration



Division

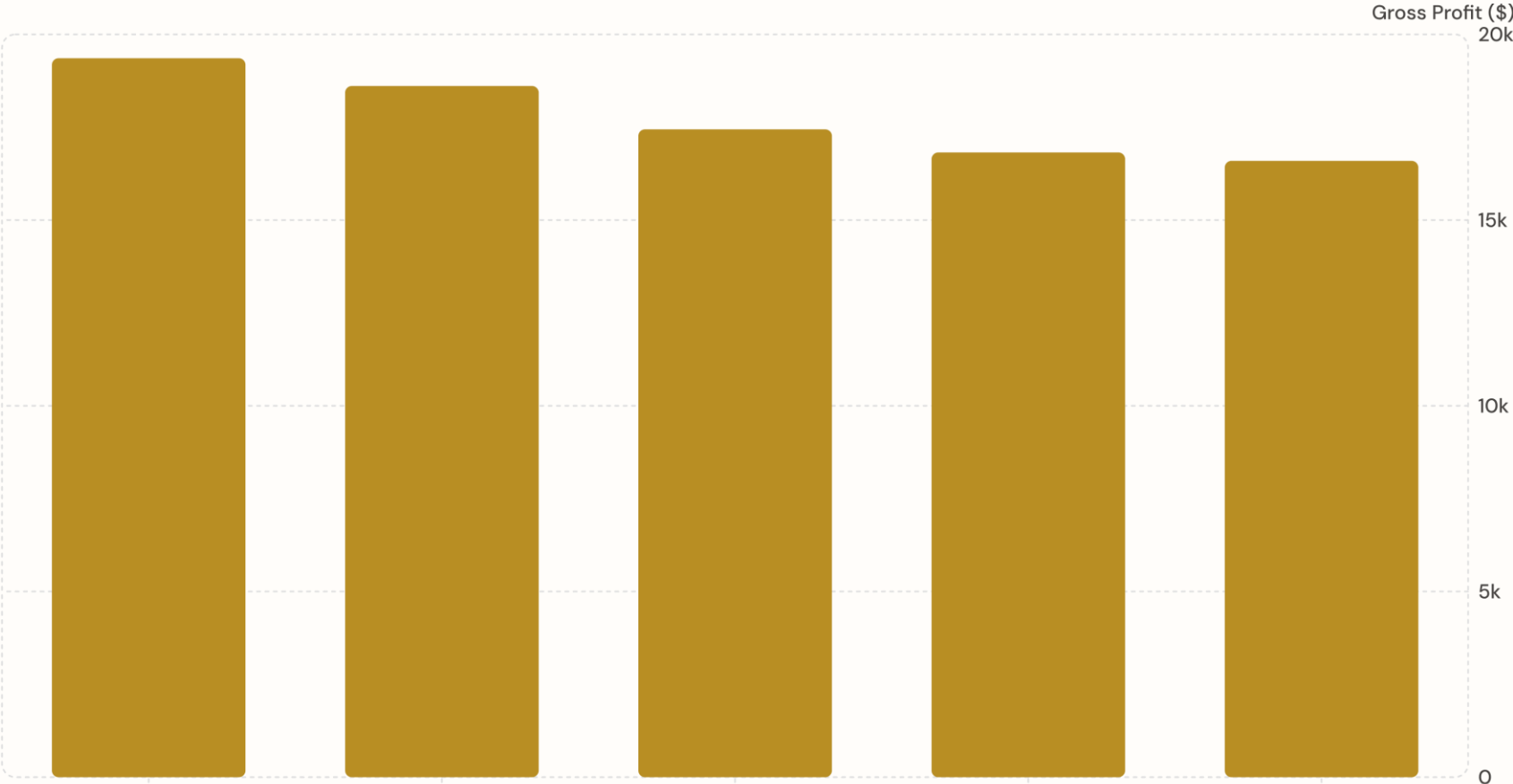
Chocolate dominates gross profit:
\$88,824.62 vs Other \$4,333.45 and Sugar
\$284.73.



Region

Pacific leads gross profit with \$30,485.94,
followed by Atlantic \$26,973.70, Interior
\$21,282.49, Gulf \$14,700.67.

Top Products — Sales, Profit & Margin





Recommendations & Next Steps

1

Product Strategy

Prioritize Wonka Bar SKUs for inventory and promotions; use margin + volume when planning.

2

Regional Actions

Replicate Pacific product mix elsewhere; review Gulf for mix/pricing improvement opportunities.

3

Profitability Controls

Monitor gross profit and margin together; set alerts for low-margin products; schedule monthly reviews.

4

Future Enhancements

Add trend analysis, YoY comparisons, customer segmentation, and automated data refresh.

📌 Overall gross profit \$93,442.80 on sales \$141,783.63 (65.91%): strong profitability baseline to guide product and regional decisions.